

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: COMPILER DESIGN

COURSE OUTCOME COVERED

CO5: *Develop code optimization and Code Generation using DAG representation with data flow analysis to produce optimized target code (BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 30

Total Marks:60

Annexure A

MOCK INTERVIEW

Code of Conduct:

I K. Sabitha (192524308) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sabitha

Annexure A

MOCK INTERVIEW

QUESTIONS

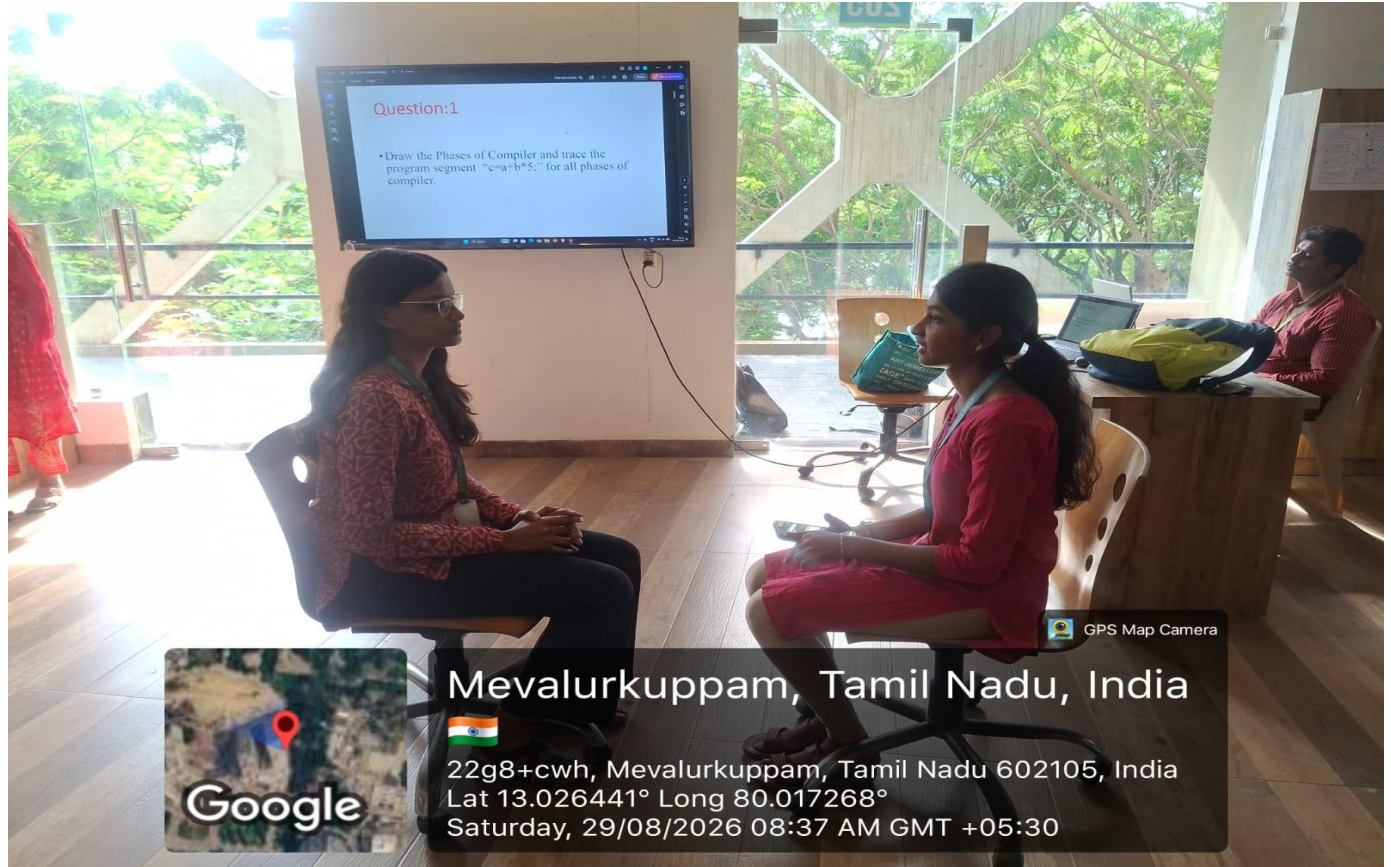
1. What is a compiler?
2. What are the different phases of a compiler?
3. What is lexical analysis?
4. What is the role of a lexical analyzer?
5. What is a token? Give an example.
6. What is a lexeme?
7. What is the difference between token and lexeme?
8. What is a pattern in lexical analysis?
9. What are keywords and identifiers?
10. What are the common errors detected by a lexical analyzer?
11. What is syntax analysis?
12. What is the role of a parser?
13. What is a parse tree?
14. What is the difference between parse tree and syntax tree?
15. What is ambiguity in grammar?
16. What is left recursion?
17. Why should left recursion be removed?
18. What is left factoring?
19. What is predictive parsing?
20. What is the dangling else problem?

21. What is FIRST set?
22. What is FOLLOW set?
23. Why are FIRST and FOLLOW sets important?
24. What is LL(1) grammar?
25. What is an LL(1) parsing table?
26. When does a conflict occur in LL(1) parsing?
27. Why is ambiguous grammar not suitable for LL(1)?
28. How does a predictive parser work?
29. What is bottom-up parsing?
30. What is shift-reduce parsing?
31. What is a handle?
32. What is handle pruning?
33. What is a shift-reduce conflict?
34. What is a reduce-reduce conflict?
35. What is operator precedence parsing?
36. What is the difference between top-down and bottom-up parsing?
37. What is LR parsing?
38. What is an augmented grammar?
39. What is closure in LR parsing?
40. What is GOTO function?
41. What is SLR parsing?
42. What is the difference between LR(0) and SLR?
43. What is semantic analysis?
44. What is a symbol table?
45. What information is stored in a symbol table?
46. What is type checking?
47. What is intermediate code generation?

48. What is three-address code?

49. What is code optimization?

50. What is peephole optimization?



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking was demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately